BUSINESS ANALYSIS

Key Performance Indicators (KPIs)

KEY PERFORMANCE INDICATORS

- **What are KPIs**

  "Key Performance Indicators (KPIs) are the critical, quantifiable measures of progress toward a desired result. They help organizations determine if their efforts are making an impact, allocate resources effectively, and focus improvements where they matter most."
- **Why is it important to have KPIs**

  "Drive Performance Improvement": KPIs highlight the metrics that are the most important for an organization.

  "Enable Data Driven Decisions": The management can measure progress on the KPIs and make decisions, e.g., resource allocation accordingly.

  "Ensure Strategic Alignment": Help align and direct organizational tasks/efforts towards measurable objectives

Video: What is a key performance indicator (KPI)?

Source: KPI.org; https://www.kpi.org/kpi-basics/

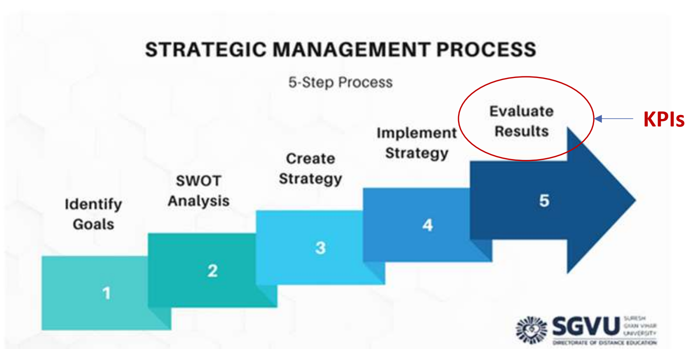
KPI – DESIRABLE CHARACTERISTICS (Kerzner, 1992: Morton et al, 1981)

- **Specific & Accurate** – KPI must reflect what we are trying to measure in an accurate way. E.g., amount of cement delivered to site is not a good indicator of number of houses constructed.
- **Measurable** – KPIs must be measurable in either quantitative or qualitative terms. Quantifiable KPIs are desirable because they can be precise, aggregated and allow for statistical analysis. However, intangible project objective may require qualitative KPIs.
- **Agreed** - KPIs should be derived in negotiation with the key stakeholders.
- **Realistic** – The KPI must be attainable at reasonable cost using appropriate collection methods
- **Timely** – KPI measurements must in obtained in a timely manner to ensure proactive action when required

KPIs AND STRATEGIC MANAGEMENT

What is Strategic Management?

Strategic management is the process of setting goals, making decisions, and taking actions to achieve long-term success. It is the big picture that guides everything a company does, from where it invests its resources to how it adapts to changes in the market.

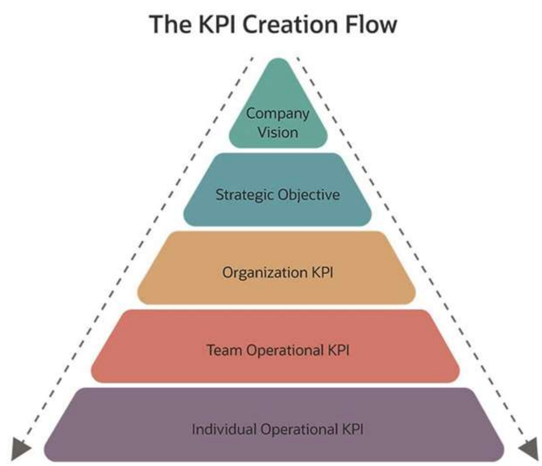


Source: Medium; https://medium.com/@sgvuguide/strategic-management-what-is-it-and-its-process-4ea4330a08b1

KPIs AND STRATEGIC MANAGEMENT

"How Are KPIs Used?

- KPIs are used to track progress toward a business goal. KPIs are created, communicated, and then tracked over a period that coincides with the underlying objective.
- Organizations typically analyze KPIs at intervals to determine whether course corrections are needed.
- KPIs may be evaluated during this process. Some KPIs may be discontinued, revised, or renewed once a goal has been reached.
- KPIs are often used for employee recognition, development, and financial rewards."



Source: ORACLE NetSuite; https://www.netsuite.com/portal/resource/articles/erp/key-performance-indicators-kpis.shtml

STRATEGIC VS. OPERATIONAL KPIs

- **Strategic KPIs** connect directly to your organization's long-term vision and highest priorities.
  - High-level indicators that help leadership gauge progress toward ambitious goals that typically span years rather than months.
  - Examples: market share growth, customer lifetime value, or return on capital employed (KPIs that reveal the organization's trajectory toward lasting success)
- **Operational KPIs** monitor the day-to-day and week-to-week activities that power your organization.
  - While seemingly tactical, well-chosen operational KPIs directly support strategic objectives
  - Examples: Production cycle time, call resolution rate, or inventory turnover (KPIs that provide vital feedback on process efficiency and effectiveness

Source: Spider Strategies; https://www.spiderstrategies.com/blog/what-is-a-kpi/

LEADING VS. LAGGING KPIs

- **Leading KPIs** Leading indicators act as early warning systems and predictors of future performance. They measure activities and trends that typically precede desired outcomes.

  Examples:
  - Sales pipeline value (all active deals) may predict future revenue
  - Employee engagement scores (e.g., survey results of employees motivation level) may forecast future turnover rates
  - Customer support ticket volume might indicate upcoming changes in satisfaction scores

- **Operational KPIs** measure results after they've occurred, confirming whether objectives were achieved; represent the outcomes of actions already taken; arrive too late for course correction on the activities they measure

  Examples:
  - Quarterly revenue
  - Customer churn rate
  - Annual profit margins

Source: Spider Strategies; https://www.spiderstrategies.com/blog/what-is-a-kpi/

OTHER CATEGORIZATION SCHEMES FOR KPIs

Quantitative vs. Qualitative KPIs

Quantitative Examples:
- 15% sales growth
- 3.2% defect rate
- $82 average order value

Qualitative Examples:
- Brand perception
- Customer satisfaction
- Employee sentiment
- Product quality evaluation

Source: Spider Strategies; https://www.spiderstrategies.com/blog/what-is-a-kpi/

OTHER CATEGORIZATION SCHEMES FOR KPIs

Financial vs. Non-Financial KPIs

Financial Examples:

- Gross profit margin
- Operating expense ratio
- Return on invested capital (ROIC)

Non-financial Examples:

- Customer satisfaction and loyalty
- Market penetration
- Innovation rates
- Process efficiency

Source: Spider Strategies; https://www.spiderstrategies.com/blog/what-is-a-kpi/

OTHER CATEGORIZATION SCHEMES FOR KPIs

Input, Process, Output, and Outcome KPIs

Input Examples:

- Budget allocation
- Staff hours
- Raw materials

Process Examples:

- Cycle time
- Error rates
- Resource utilization

Process Examples:

- Units manufactured
- Services delivered
- Tasks completed

Source: Spider Strategies; https://www.spiderstrategies.com/blog/what-is-a-kpi/

KPIs AND STRATEGIC MANAGEMENT

Using the Balanced Scorecard (Robert Kaplan & David Norton, 1992) **for designing KPIs**

Customer perspective: KPIs that focus on customer interactions and their views on product quality and performance

Financial perspective: KPIs that examine profitability, liquidity, risk, and financial efficiency e.g., sales made in a period

Growth perspective: KPIs that measure returns on investment in equipment, research, and staff

Internal Process perspective: KPIs that focus on the efficiency of specific processes.

Video: The Balanced Scorecard explained

Source: ORACLE NetSuite; https://www.netsuite.com/portal/resource/articles/erp/key-performance-indicators-kpis.shtml
EPM: https://www.youtube.com/watch?v=C0JZdyb6hZE

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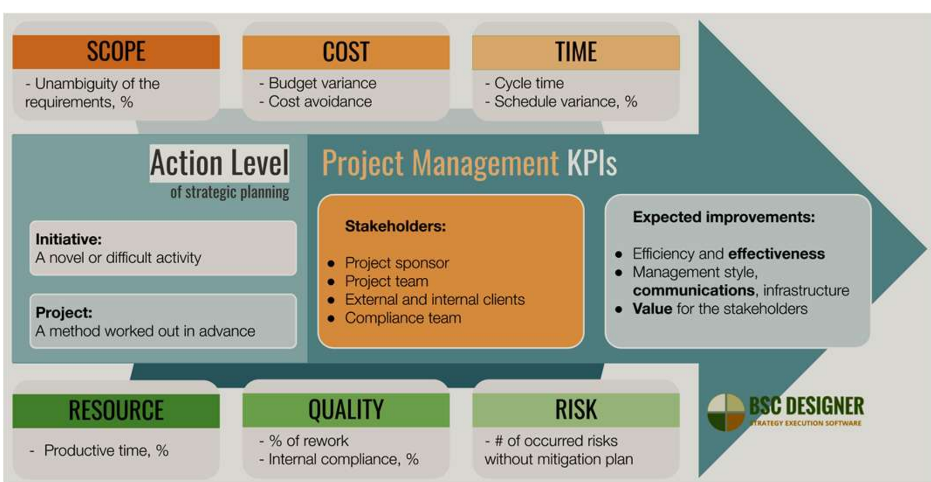
KPIs AND PROJECT MANAGEMENT



Source: BSC Designer; https://bscdesigner.com/project-management-kpis.htm

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EXAMPLE: MELISSA NEEDS A NEW CAR (adapted from a case project by Dept. of Construction Mngt, Curtin University)

- **Project Rationale**

  Melissa's "Old Bomb" model had been detected with several issues requiring extensive repairs with cost in excess of $ xxxx. Given the market value of the Old Bomb and potential for additional expense in the short term, the client has decided not to proceed with repairs to the vehicle. In addition, the Old Bomb no longer meets family and recreational requirements. The client has requested that a replacement vehicle be researched, selected and purchased.

- **Strategic contribution**

  The "Melissa Needs a New Car" project will contribute to the long term goals of Happiness, Self-esteem and Financial Wellbeing.

- **Project Purpose**

  The sale of the existing vehicle and replacement with a new and improved model will enable:

  1. Decreased on-going vehicle service costs.
  2. Decreased occurrence of vehicle breakdown and the stress associated with it.
  3. Lowered "embarrassment factor" of driving in a clapped out vehicle.
  4. Increased safety for the driver and passengers while driving in the vehicle.
  5. Improved ease of transport of recreational equipment, shopping, hardware.
  6. Improved ability to transport of friends and family.
  7. Increased enjoyment of driving.

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Key Performance Indicators – Benefits

Key Performance Indicators (KPIs) have been set for each of the Project Purposes

BENEFITS	KPIs
Decreased on-going vehicle costs	Service costs decreased by 50% over five years
Decreased chance of vehicle breakdown	Number of breakdowns reduced to one per year over five years.
Lowered "embarrassment factor"	Number of incidences of parking vehicle far away from destination and walking the rest of the way reduced to zero upon purchase and for five years from purchase.
Increased safety while driving	Number of safety features in the new vehicle double the number of safety features in the old vehicle at time of purchase.
Improved ease of transport of recreational equipment, shopping, hardware	Space available for storage and transport of items doubled at time of purchase.
Improved ability to transport friends and family	Frequency of "lifts" given to friends and family by 200% over five years.
Increased enjoyment of driving.	Score in "Driving Enjoyment" survey increased above threshold score of 8/10 in the first year and maintained over five years. (Note – "Driving Enjoyment" survey is a self administered survey).

ESSENTIAL READINGS

BSC Designer: “Project Management KPIs”, available at https://bscdesigner.com/project-management-kpis.htm

EPM: “The Balanced Scorecard explained with examples”, available at https://www.youtube.com/watch?v=C0JZdyb6hZE

KPI.ORG: “What is a key performance indicator”, available at https://www.kpi.org/kpi-basics/

Medium: “Strategic management – what is it?”, available at https://medium.com/@sgvuguide/strategic-management-what-is-it-and-its-process-4ea4330a08b1

Oracle NetSuite: “What is a key performance indicator? A guide to KPIs”, available at https://www.netsuite.com/portal/resource/articles/erp/key-performance-indicators-kpis.shtml

Spider Strategies: “What Is a KPI? Complete guide to key performance indicators”, available at https://www.spiderstrategies.com/blog/what-is-a-kpi/

REFERENCE BOOKS OPTIONS

- **BUSINESS ANALYSIS BODY OF KNOWLEDGE (BABOK);** IIBA Bookstore: https://www.iiba.org/career-resources/a-business-analysis-professionals-foundation-for-success/babok/
- **BUSINESS ANALYSIS (Fourth Edition)** by Debra Paul (Author), James Cadle (Author), Malcolm Eva (Author), Craig Rollason (Author), Jonathan Hunsley (Author)
- **BUSINESS ANALYSIS TECHNIQUES: 123 ESSENTIAL TOOLS FOR SUCCESS (3RD Edition)** by James Cadle (Author), Debra Paul (Author), Jonathan Hunsley (Author), Adrian Reed (Author)